PHYSICS TEST

DATE: May 30, 2026 **TIME: 1 HOURS** **TOTAL MARKS: 25**

STUDENT NAME: ________________________ ROLL NO: ____________ SEC: ______

GENERAL INSTRUCTIONS:

1. Read all questions carefully before answering.
2. Write your answers neatly in the space provided.
3. Verify all code syntax or reasoning steps where requested.

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer all questions. Each question carries 2 marks. Choose the most appropriate option.

1. Which of the following is a vector quantity? **[Easy]** **[2 Marks]**

   (a) Speed
   (b) Distance
   (c) Mass
   (d) Velocity

2. What is the SI unit of thermodynamic temperature? **[Easy]** **[2 Marks]**

   (a) Celsius
   (b) Kelvin
   (c) Fahrenheit
   (d) Joule

3. An object is placed at the center of curvature of a concave mirror. The image formed by the mirror is: **[Moderate]** **[2 Marks]**

   (a) Real, inverted and same size
   (b) Virtual, erect and magnified
   (c) Real, inverted and diminished
   (d) Virtual, erect and diminished

4. If the resistance of a conductor is doubled while keeping the potential difference across its ends constant, the current flowing through it becomes: **[Moderate]** **[2 Marks]**

   (a) Doubled
   (b) Halved
   (c) Four times
   (d) Unchanged

5. Which of the following phenomena provides the most direct evidence for the wave nature of light? **[Hard]** **[2 Marks]**

   (a) Photoelectric effect
   (b) Blackbody radiation
   (c) Interference
   (d) Compton scattering

SECTION B: SHORT ANSWER QUESTIONS

Answer the following question in detail. This question carries 5 marks.

6. State Newton's Second Law of Motion and explain how it leads to the concept of inertial mass. **[Moderate]** **[5 Marks]**

SECTION C: DIAGRAM/GRAPH-BASED QUESTIONS

Analyze the described graphical scenario and answer the questions. This question carries 5 marks.

7. A velocity-time graph of a moving car shows a straight line starting from rest with a positive slope for 5 seconds, followed by a horizontal line for 10 seconds, and finally a straight line with a negative slope coming to rest in 5 seconds. (a) Describe the motion of the car in each of the three phases. (b) Explain how you would calculate the total distance traveled by the car from this graph. **[Hard]** **[5 Marks]**

SECTION D: NUMERICAL PROBLEMS

Solve the following numerical problem showing all steps and formulas used. This question carries 5 marks.

8. An electric heater is rated 220 V, 1100 W. Calculate: (a) The resistance of the heater element when in use. (b) The current flowing through the heater. **[Moderate]** **[5 Marks]**